Modeling and Forecasting VN-Index Volatility: A Financial Econometric Evaluation of GARCH-Family, HAR, Neural, and Hybrid Approaches

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Abstract—**This paper studies one-step-ahead daily VN-Index volatility forecasting using CafeF OHLC and trading-activity data. The target is next-day squared percentage log return, and corrected-context neural retraining expands the unified primary out-of-sample window to 745 common target dates from 2023-01-04 to 2025-12-31. The study compares historical and rolling variance baselines, EWMA/RiskMetrics, HAR-type models, GARCH(1,1), ARIMA-GARCH, validation-selected GARCH-family specifications, LSTM, hybrid neural forecasts, calibration, and forecast combinations. Descriptive diagnostics show negatively skewed, heavy-tailed returns, and ARCH-LM tests strongly reject homoskedasticity, motivating conditional-variance models. On the corrected common window, AdvGARCH-BestAsymmetric obtains QLIKE 1.0300, the primary validation-selected AdvGARCH-BestQLIKE obtains 1.0325, HAR-Parkinson obtains 1.0345, and GARCH(1,1) obtains 1.0630. Corrected neural and hybrid forecasts do not outperform these econometric benchmarks on average QLIKE, although validation-frozen combinations improve some spike-regime diagnostics. A regime-aware combination has test QLIKE 1.0643 and spike-day QLIKE 7.35, improving over a static combination but not over the best average-loss econometric models. Risk diagnostics qualify the average-loss ranking. Under a common-Normal zero-mean variance-to-VaR mapping, GARCH(1,1) has 16 one-percent violations out of 745 observations, against 7.45 expected, with Kupiec $p = 0.0063$. Thus good QLIKE does not by itself guarantee strict-tail calibration.**

Index Terms—**VN-Index, volatility forecasting, GARCH, EGARCH, LSTM, hybrid forecasting, QLIKE, VaR backtesting, range-based volatility.**

I. Introduction

Volatility forecasting is a practical market-risk problem. For an equity index, a one-step-ahead conditional-variance forecast summarizes the expected scale of the next return shock and feeds into risk limits, margin decisions, portfolio monitoring, and Value-at-Risk (VaR). A model can look accurate on average while still being too low when volatility spikes, which is exactly when risk forecasts matter most.

The VN-Index is a useful Vietnamese equity-market case study. It is an emerging-market index with visible volatility clustering, sharp drawdowns, and changing trading conditions, but without the routine high-frequency realized-volatility data often used in developed-market volatility studies. Daily open, high, low, close, volume, and trading-value data therefore create a realistic empirical setting: forecasts must be made from daily information, and latent volatility must be evaluated through imperfect observable proxies.

This paper is positioned as a reproducible empirical financial econometrics study, not as a new forecasting algorithm. The main question is whether daily VN-Index volatility can be forecast reliably using standard financial econometric models, and whether neural or hybrid alternatives provide robust gains once model selection, proxy uncertainty, and tail-risk diagnostics are considered. The main target is next-day squared percentage log return, and the primary comparison uses a leakage-safe common test window with identical dates for all primary models.

The contribution is fivefold:

1) A reproducible, leakage-safe out-of-sample comparison of daily VN-Index volatility forecasts on a corrected 745-observation common test window.
2) Econometric evidence on VN-Index return non-normality, volatility clustering, ARCH effects, and conditional-variance persistence.
3) A validation-based comparison of EWMA, HAR, GARCH-family, corrected-context neural, hybrid, calibrated, and combined volatility forecasts.
4) Proxy-robustness, spike-underprediction, refit-robustness, common-Normal VaR/ES, and model-specific Normal GARCH VaR/ES diagnostics that connect average forecast accuracy to practical risk interpretation.
5) A controlled validation-frozen regime-aware combination that tests whether corrected neural forecasts add useful high-volatility information without test-period tuning.

The verified evidence supports a conservative conclusion. Selected GARCH-family specifications and HAR-Parkinson form the strongest group of daily VN-Index volatility forecasts under the unified common-window protocol. The results are economically consistent with persistent conditional variance and useful OHLC range information. The implemented neural and hybrid forecasts may contain complementary information, but they do not robustly dominate the strongest econometric benchmarks. Tail-risk diagnostics further show that good average QLIKE does not automatically imply adequate one-percent VaR coverage under a common-Normal mapping.

II. Related Work

ARCH and GARCH models provide the classical foundation for conditional volatility forecasting. Engle introduced ARCH to model time-varying conditional variance, and Bollerslev generalized the recursion to include lagged conditional variance [1], [2]. The empirical appeal of GARCH(1,1) is not only parsimony. Large-scale forecast comparisons have shown that it can be difficult to beat in practical volatility forecasting applications [3].

Mean dynamics are often modeled before conditional variance. The Box-Jenkins ARIMA framework captures serial dependence in univariate time series [4]. In financial returns, the conditional mean is usually small relative to the conditional variance, but an ARMA or ARIMA mean model can still be paired with GARCH to test whether residual variance forecasts improve after filtering predictable mean components.

EWMA/RiskMetrics is a standard industry benchmark for volatility and VaR forecasting because it provides a transparent exponentially weighted recursion with very few tuning choices [5]. Its inclusion matters empirically: if a sophisticated model does not clearly improve on EWMA under volatility-specific loss and tail diagnostics, the additional complexity is hard to justify. **EWMA is the minimum credible industry benchmark for this setting.**

HAR-type models provide another important benchmark class. The heterogeneous autoregressive realized-volatility model captures persistence at daily, weekly, and monthly horizons through a simple linear structure [6]. Even when high-frequency realized volatility is unavailable, HAR-style regressions using squared-return or OHLC range proxies are useful because they test whether multi-horizon volatility persistence explains daily variance better than naive rolling volatility.

GARCH-family extensions address stylized features that a symmetric GARCH(1,1) recursion may miss. EGARCH models allow asymmetric volatility responses and avoid some positivity restrictions [7]. GJR-type models capture sign-dependent shock effects [8]. APARCH and FIGARCH specifications allow richer power transformations and long-memory volatility dynamics [9], [10]. These extensions motivate the broad specification search evaluated in this paper.

Neural sequence models offer a different forecasting premise. LSTM networks were designed to learn long-range sequential dependencies through gated recurrent states [11]. In volatility forecasting, LSTMs can use lagged returns, volume, rolling volatility, and econometric forecasts as supervised features. Hybrid econometric-neural designs are therefore useful empirical candidates, but **their performance must be judged against strong volatility-specific baselines rather than weak naive alternatives.**

Forecast evaluation is central because latent volatility is not directly observed. Squared daily returns are available and simple, but they are noisy proxies for conditional variance. QLIKE is widely used because it remains attractive when forecast comparisons rely on imperfect volatility proxies and because it evaluates forecasts on the variance scale [12]. High-frequency realized volatility can provide richer targets when intraday data are available [13], but daily OHLC estimators are also useful. Parkinson, Garman-Klass, Rogers-Satchell, and Yang-Zhang estimators exploit high, low, open, and close information to obtain alternative volatility proxies [14]–[17].

Statistical forecast comparisons require care when many models are examined. Diebold-Mariano tests compare predictive accuracy under a specified loss difference process [18]. Holm correction controls family-wise error rates across multiple tests [19]. The model confidence set (MCS) framework provides a complementary way to identify a set of models not statistically eliminated under a loss criterion [20]. VaR backtesting adds a risk-management view through unconditional and conditional coverage tests [21], [22]. **Multiplicity-aware inference is necessary before claiming single-model superiority.**

TABLE I
Chronological data splits and common evaluation window. After corrected-context neural retraining, the primary common-window comparison aligns all primary forecasts on 745 target dates from 2023-01-04 to 2025-12-31.

Dataset	Rows	Origin date range	Target date range
Prepared full	3989	2010-01-04 to 2025-12-31	2010-01-05 to 2025-12-31
Model-ready	3968	2010-02-01 to 2025-12-30	2010-02-02 to 2025-12-31
Train	2471	2010-02-01 to 2019-12-30	2010-02-02 to 2019-12-31
Validation	750	2020-01-02 to 2022-12-29	2020-01-03 to 2022-12-30
Test	745	2023-01-03 to 2025-12-30	2023-01-04 to 2025-12-31
Primary common test	745	2023-01-03 to 2025-12-30	2023-01-04 to 2025-12-31

III. Data and Econometric Motivation

A. Daily VN-Index Data

The empirical sample uses CafeF VNINDEX daily observations from 2010 through 2025. The cleaned data include open, high, low, close, matched trading volume, and matched trading value. Table I reports the chronological splits. The model-ready sample contains 3968 rows from 2010-02-01 to 2025-12-30 after lagged-return, rolling-volatility, and next-day-target requirements are imposed.

The split is chronological. Training observations end in 2019, validation covers 2020–2022, and the test period begins in 2023. After corrected-context neural retraining, the authoritative primary leaderboard uses the common intersection of all primary prediction files: forecast origins from 2023-01-03 to 2025-12-30 and target dates from 2023-01-04 to 2025-12-31, with $N = 745$ for every primary model.

B. Return and Target Definitions

Let C_t denote the VN-Index close on trading day t. Percentage log return is

$$r_t = 100 \log \left(\frac{C_t}{C_{t-1}} \right), \tag{1}$$

TABLE II
DESCRIPTIVE STATISTICS FOR DAILY VN-INDEX PERCENTAGE LOG RETURNS.

Statistic	Value
Observations	3968
Mean	0.032738
Std. dev.	1.174558
Minimum	-6.910236
Maximum	6.546905
Skewness	-0.781445
Excess kurtosis	3.979439
Jarque-Bera stat.	3013.172
Jarque-Bera p-value	0

TABLE III
RETURN-DEPENDENCE AND ARCH-EFFECT DIAGNOSTICS.

Test	Series	Lag	Statistic	p-value
Ljung-Box	returns	5	41.839	6.35e-08
Ljung-Box	returns	10	49.343	3.52e-07
Ljung-Box	returns	20	60.711	5.53e-06
Ljung-Box	squared_returns	5	794.846	1.51e-169
Ljung-Box	squared_returns	10	994.918	2.33e-207
Ljung-Box	squared_returns	20	1081.398	1.67e-216
ARCH-LM	demeaned_returns	5	455.264	3.6e-96
ARCH-LM	demeaned_returns	10	480.760	5.69e-97
ARCH-LM	demeaned_returns	20	496.552	1.53e-92

and the daily squared-return volatility proxy is

$$v_t = r_t^2. \tag{2}$$

The supervised one-step target is

$$y_t = v_{t+1} = r_{t+1}^2. \tag{3}$$

Thus a forecast issued at origin date t uses information available through t to predict the variance proxy for target date $t+1$. Returns are measured in percentage units, so v_t and y_t are percentage-squared variance proxies.

Squared daily return is observable and simple, but noisy. A single close-to-close return can be small even when intraday variation is high, and a single jump can dominate the proxy. For this reason, the paper uses QLIKE as the main loss and reports robustness checks with daily OHLC range-based volatility proxies.

C. Descriptive Return Evidence

Table II shows that VN-Index daily returns are not approximately Gaussian. The mean return is small relative to the standard deviation, while skewness is negative and excess kurtosis is high. The Jarque-Bera statistic strongly rejects normality. This matters for volatility and VaR because extreme observations are more frequent than a Normal model would suggest, and downside shocks are economically important for market-risk applications.

D. Volatility Clustering and ARCH Effects

Table III provides formal motivation for conditional-variance modeling. Ljung-Box tests on squared returns and ARCH-LM tests reject the absence of volatility dependence at lags 5, 10, and 20. Raw return dependence is weaker and economically less central than squared-return dependence.

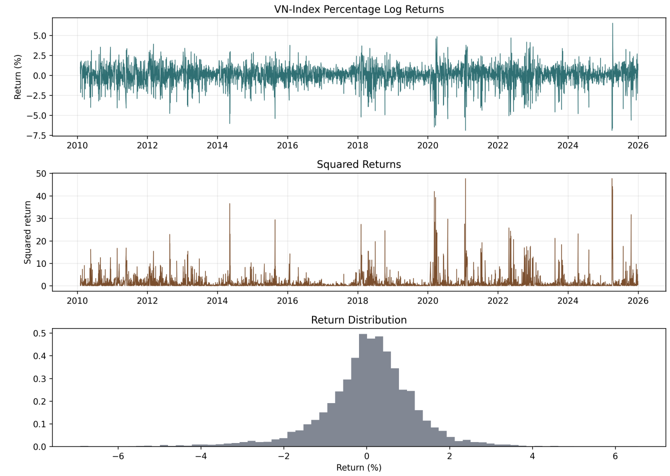

Fig. 1. VN-Index returns, squared returns, and return distribution. The time series shows volatility clustering and occasional large shocks.

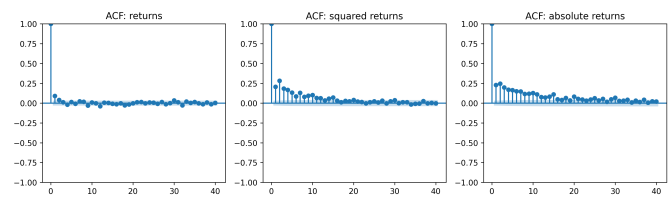

Fig. 2. ACF diagnostics for returns, squared returns, and absolute returns. Squared and absolute returns display more persistent dependence than raw returns.

This is the standard econometric pattern behind ARCH and GARCH models: conditional means are hard to forecast, but conditional variance is persistent.

E. OHLC Volatility Proxies

The daily proxy robustness analysis uses close-to-close squared return, Parkinson, Garman-Klass, and Rogers-Satchell variance proxies as same-horizon daily alternatives. Rolling Yang-Zhang 5-, 10-, and 20-day proxies are retained only as smoothed sensitivity checks because they are overlapping measurements and not identical to the one-day-ahead squared-return target. The main paper therefore emphasizes the daily proxy results and treats rolling proxy rankings as appendix-level sensitivity evidence.

IV. FORECASTING MODELS AND EXPERIMENTAL PROTOCOL

A. Benchmark Models

The benchmark set begins with simple variance forecasts. The historical mean predicts the training-sample average squared return. RollingVol-5, RollingVol-10, and RollingVol-20 use trailing rolling volatility, squared to produce variance forecasts. These baselines are intentionally simple, but they are useful because volatility persistence can make recent realized variation competitive.

The EWMA/RiskMetrics model follows

$$\sigma_{t+1}^2 = \lambda \sigma_t^2 + (1-\lambda) r_t^2, \tag{4}$$

TABLE IV
ADVANCED GARCH FAILURE TAXONOMY. CATEGORIES SUM EXACTLY TO THE 576 FAILED CANDIDATES.

Failure category	Count	Example
invalid_forecast_start_for_ar_mean_lag	432	AR(2), ARCH(1), normal
nonfinite_forecast	144	AR(1), ARCH(1), normal

where r_t^2 is the current squared percentage return. The decay parameter is selected by validation QLIKE from $\lambda \in \{0.90, 0.94, 0.97, 0.99\}$, and the selected model is EWMA($\lambda = 0.90$).

HAR-type models approximate heterogeneous volatility persistence with daily, weekly, and monthly components:

$$y_{t+1} = \beta_0 + \beta_d RV_{d,t} + \beta_w RV_{w,t} + \beta_m RV_{m,t} + \varepsilon_{t+1}. \quad (5)$$

HAR-SquaredReturn uses squared-return components. HAR-Parkinson replaces them with daily and averaged Parkinson range-variance components and is selected as the primary HAR variant by validation QLIKE.

B. GARCH-Family Models

The standard GARCH(1,1) benchmark is

$$r_t = \mu + \epsilon_t, \qquad \epsilon_t = \sigma_t z_t, \quad (6)$$
$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2. \quad (7)$$

Parameters are estimated on the training split for the static protocol. Validation and test forecasts use fixed estimated parameters while recursively updating the volatility state with returns observed before each forecast origin.

ARIMA-GARCH first models return mean dynamics using a training-AIC ARIMA selection over $p, q \in \{0, 1, 2, 3\}$ with $d = 0$, then fits GARCH(1,1) to the residuals. The original benchmark selects ARIMA(0,0,3).

The advanced GARCH-family search evaluates direct ARCH-family and two-step ARIMA-GARCH candidates. It includes ARCH, GARCH, EGARCH, GJR-GARCH, APARCH, and FIGARCH-type specifications with Normal, Student-t, skewed-t, and GED innovations where supported. Candidate selection uses validation QLIKE only. The primary advanced model is AdvGARCH-BestQLIKE, an ARIMA(3,0,3)-EGARCH(1,1,2)-Normal specification. AdvGARCH-BestAsymmetric is reported as a category-selected asymmetric comparator, not as the primary winner merely because its test QLIKE is slightly lower.

The search attempts 3,168 candidates, with 2,592 successful fits and 576 failures. Table IV reconciles the failures: 432 are invalid forecast-start failures for AR-mean lag specifications and 144 are non-finite forecast failures. This audit is important because large specification searches can look precise while hiding numerical fragility.

C. Neural, Hybrid, Calibration, and Combination Models

The base LSTM uses length-20 sequences of lagged market and volatility features, including returns, squared returns, absolute returns, rolling volatility, volume, and trading value. The ARIMA-GARCH-LSTM hybrid augments these inputs with econometric variance and residual-related features. Tuned neural variants include log-target training and QLIKE-style loss training; selection is by validation QLIKE within the neural group.

Calibration methods include multiplicative QLIKE scaling, mean matching, nonnegative affine calibration, and isotonic calibration fitted on validation data only. Forecast combinations include equal-weight means, medians, two-model convex combinations, and stacking weights selected by validation QLIKE. Test-best combinations are treated as exploratory and are not used as primary evidence.

The neural sequence-construction code was corrected so validation and test forecasts may use valid pre-split historical context without target leakage. All neural and hybrid forecasts used in the final tables were retrained under the versioned namespace `outputs/neural_corrected_context_v2/`. The legacy pre-correction artifacts are preserved under `outputs/neural_legacy_pre_context_fix/`. Training uses seed 42, a sequence length of 20, training-only imputation and scaling, validation-monitored early stopping, and TensorFlow deterministic options where available. Corrected calibration and forecast-combination artifacts are regenerated under `outputs/advanced_corrected_context_v2/`; calibration parameters and combination weights are fitted on validation data only.

D. Evaluation Protocol

For aligned observations $\{y_t, \hat{y}_t\}_{t=1}^n$, the evaluation uses

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2}, \quad (8)$$

$$\text{MAE} = \frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t|, \quad (9)$$

$$\text{QLIKE} = \frac{1}{n}\sum_{t=1}^{n}\left(\log(\hat{y}_t) + \frac{y_t}{\hat{y}_t}\right). \quad (10)$$

Predicted variances are clipped at $\epsilon = 10^{-8}$ before QLIKE computation. QLIKE is central because severe variance underprediction is penalized through $y_t/\hat{y}_t$.

Paired comparisons align observations by origin and target date, not by row order. Diebold-Mariano-style tests use loss differences relative to GARCH(1,1), and Holm correction is applied within clearly defined loss families. Regime and spike diagnostics use train/validation actual-variance thresholds to avoid selecting spike thresholds from test outcomes. The standardized risk layer maps every variance forecast to lower-tail return VaR and ES using a common zero-mean Normal distribution at 5%, 2.5%, and 1%. A second layer reports model-specific Normal VaR/ES for selected GARCH-family models whose innovation distribution is verified as Normal. Student-t, GED, and skewed-t ES are not reported unless parameter extraction and ES formulas are verified.

E. Reproducibility Statement

The repository is a git repository on branch `master`, with remote https://github.com/trantuan1701/TSA_ARIMA-GARCH-LSTM.git verified by `git ls-remote`. The reproduction snapshot should be the release commit or tag that contains the packaged root `paper.pdf`, the root `README.md`, and the final artifact manifest. Raw CafeF data are available in the local course-submission repository under `data/raw/`, but no redistribution license is present; a public release should either add a data-license statement or provide download/preprocessing instructions. The main reproduction commands are `make neural-corrected`, `make advanced-corrected-combinations`, `PYTHONDONTWRITEBYTECODE=1 make final-artifacts`, `PYTHONDONTWRITEBYTECODE=1 pytest -q`, and `make -C paper all`. Artifact hashes and provenance are recorded in `outputs/artifact_manifest.csv` and `outputs/artifact_manifest.json`.

F. Expanding-Window Refit Robustness

The refit robustness experiment does not rerun the full advanced search. It fixes the validation-selected specifications and compares the static train-only forecast protocol with an expanding-window monthly refit protocol for GARCH(1,1), ARIMA-GARCH, EWMA($\lambda = 0.90$), HAR-Parkinson, AdvGARCH-BestQLIKE, and AdvGARCH-BestAsymmetric. At each monthly refit date, parameters are estimated using observations available before that month, and one-step forecasts are generated for that month. Failed refits are recorded explicitly rather than silently dropped.

V. Empirical Results

A. Primary Common-Window Leaderboard

Table V is the authoritative primary leaderboard after corrected-context neural retraining. Every listed model is evaluated on the same 745 target dates from 2023-01-04 to 2025-12-31 and the same next-day squared percentage log-return target. The earlier 726-date window is retained only as a legacy comparability check because it was driven by pre-correction neural sequence coverage.

The strongest group still consists of selected GARCH-family specifications and HAR-Parkinson. AdvGARCH-BestAsymmetric has the lowest corrected-window QLIKE, 1.029992, but it is a category-selected asymmetric comparator. The primary validation-selected advanced model, AdvGARCH-BestQLIKE, is very close at 1.032548. HAR-Parkinson follows with QLIKE 1.034521, ahead of HAR-SquaredReturn at 1.047975, GARCH(1,1) at 1.062958, and ARIMA-GARCH at 1.065995.

The corrected neural and hybrid models do not robustly improve on these econometric benchmarks. The base LSTM has QLIKE 1.163862 and the ARIMA-GARCH-LSTM has QLIKE 1.172986. The best validation-selected calibrated neural forecast in the primary inventory improves to 1.126931, but remains behind the strongest GARCH-family and HAR forecasts. This is not evidence that neural volatility models generally fail; it shows that the implemented neural and hybrid specifications do not dominate on average QLIKE in this daily VN-Index setting.

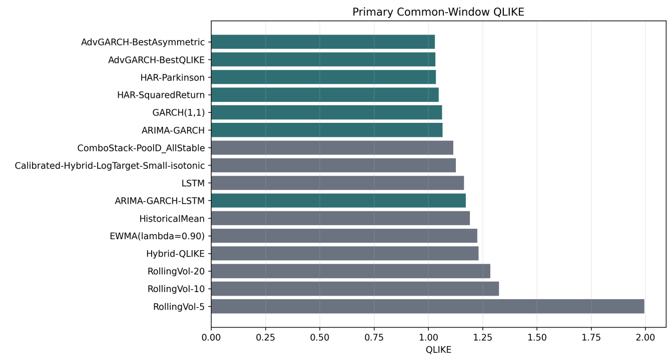

Fig. 3. Primary common-window QLIKE for the corrected 745-observation leaderboard. Lower is better.

Table VI documents the artifact change directly. Corrected-context neural predictions recover the first 19 test origins that were absent from the legacy sequence-local artifacts. The corrected neural QLIKE values are close to, but not materially better than, the legacy-window values, so the main conclusion does not depend on the sequence-coverage correction.

B. Parameter and Residual Diagnostics

Table VII gives the post-estimation financial econometric diagnostics. For GARCH(1,1), $\alpha = 0.1257$, $\beta = 0.8476$, and $\alpha + \beta = 0.9733$. The persistence is below one, so the unconditional variance and shock half-life are technically defined; the estimated half-life is about 25.6 trading days. This supports the economic interpretation that volatility shocks to the VN-Index decay slowly.

The two EGARCH specifications also display high log-variance persistence, with beta-sum measures near 0.95. Both AdvGARCH-BestQLIKE and AdvGARCH-BestAsymmetric have negative and statistically significant $\gamma[1]$ estimates at conventional levels. Under the EGARCH sign convention, this is consistent with negative shocks raising future volatility more than positive shocks of comparable standardized magnitude. The evidence supports an asymmetric volatility response, but it should still be read cautiously because the models are selected from a broad specification search.

Table VIII shows that squared standardized residual dependence is largely removed for the selected GARCH-family models. Ljung-Box tests on squared standardized residuals and ARCH-LM tests do not reject at lags 5, 10, or 20. This contrasts with the strong ARCH-LM rejections in raw returns and indicates that the models capture much of the conditional heteroskedasticity. Standardized residuals remain non-normal, however, so heavy tails and tail-risk calibration remain important.

C. Statistical Comparisons

The common-window Diebold-Mariano outputs are intentionally interpreted conservatively. Against GARCH(1,1),

TABLE V

PRIMARY COMMON-WINDOW TEST RESULTS. ALL MODELS USE TARGET DATES 2023-01-04 TO 2025-12-31 AND THE SAME NEXT-DAY SQUARED PERCENTAGE LOG-RETURN TARGET. SELECTION ROLES ARE SHORTENED; FULL PROVENANCE IS IN OUTPUTS/FINAL/PRIMARY_MODEL_INVENTORY.CSV.

Rank	Model	Selection role	N	QLIKE	RMSE	MAE	Pred./Actual
1	AdvGARCH-BestAsymmetric	Asymmetric comparator	745	1.029992	3.543267	1.321160	0.938
2	AdvGARCH-BestQLIKE	Validation AdvGARCH	745	1.032548	3.561391	1.324605	0.936
3	HAR-Parkinson	Validation HAR	745	1.034521	3.597062	1.358346	0.976
4	HAR-SquaredReturn	HAR comparator	745	1.047975	3.568915	1.364458	1.000
5	GARCH(1,1)	Canonical GARCH	745	1.062958	3.628230	1.407866	1.036
6	ARIMA-GARCH	AIC ARIMA-GARCH	745	1.065995	3.651122	1.414546	1.032
7	ComboStack-PoolD_AllStable	Validation combo	745	1.114856	3.632614	1.658267	1.378
8	Calibrated-Hybrid-LogTarget-Small-isotonic	Calibrated neural	745	1.126931	3.645180	1.635585	1.340
9	LSTM	Neural baseline	745	1.163862	3.742718	1.643025	1.299
10	ARIMA-GARCH-LSTM	Hybrid baseline	745	1.172986	3.738682	1.726914	1.411
11	HistoricalMean	Fixed baseline	745	1.191049	3.769834	1.447045	0.990
12	EWMA(lambda=0.90)	Validation EWMA	745	1.225435	3.696192	1.429298	1.019
13	Hybrid-QLIKE	Tuned hybrid	745	1.231908	3.786362	1.841543	1.518
14	RollingVol-20	Fixed baseline	745	1.285347	3.875261	1.476067	1.014
15	RollingVol-10	Fixed baseline	745	1.324963	3.952645	1.496788	1.019
16	RollingVol-5	Fixed baseline	745	1.994647	3.987033	1.504270	0.998

TABLE VI

LEGACY VERSUS CORRECTED-CONTEXT NEURAL TEST RESULTS.

model	artifact_version	n_obs	qlike	rmse	mae
ARIMA-GARCH-LSTM	corrected_context_v2	745	1.172986	3.738682	1.726914
ARIMA-GARCH-LSTM	legacy_pre_context_fix	726	1.168609	3.763253	1.720113
Hybrid-QLIKE	corrected_context_v2	745	1.231908	3.786362	1.841543
Hybrid-QLIKE	legacy_pre_context_fix	726	1.230508	3.807296	1.862578
LSTM	corrected_context_v2	745	1.163862	3.742718	1.643025
LSTM	legacy_pre_context_fix	726	1.161001	3.769716	1.639734
LSTM-QLIKE	corrected_context_v2	745	1.187248	3.754504	1.713801
LSTM-QLIKE	legacy_pre_context_fix	726	1.185876	3.782485	1.714073

TABLE VII

SELECTED GARCH-FAMILY PARAMETER DIAGNOSTICS. PERSISTENCE AND HALF-LIFE ARE REPORTED ONLY WHERE THE FORMULA IS TECHNICALLY APPROPRIATE.

Model	Variance equation	Parameter	Estimate	Std. err.	p-value	Persistence	Validity
GARCH(1,1)	GARCH(1,1)	mu	0.0456	0.0204	0.0256	0.9733	stationary
GARCH(1,1)	GARCH(1,1)	omega	0.0374	0.0163	0.022	0.9733	stationary
GARCH(1,1)	GARCH(1,1)	alpha[1]	0.1257	0.0212	3.16e-09	0.9733	stationary
GARCH(1,1)	GARCH(1,1)	beta[1]	0.8476	0.0303	1.61e-172	0.9733	stationary
AdvGARCH-BestQLIKE	EGARCH(1,1,2)	omega	0.0063	0.0057	0.273	0.9503	EGARCH persist.
AdvGARCH-BestQLIKE	EGARCH(1,1,2)	alpha[1]	0.2380	0.0271	1.84e-18	0.9503	EGARCH persist.
AdvGARCH-BestQLIKE	EGARCH(1,1,2)	gamma[1]	-0.0440	0.0186	0.0178	0.9503	EGARCH persist.
AdvGARCH-BestQLIKE	EGARCH(1,1,2)	beta[1]	0.9503	0.1492	1.91e-10	0.9503	EGARCH persist.
AdvGARCH-BestQLIKE	EGARCH(1,1,2)	beta[2]	0.0000	0.1536	1	0.9503	EGARCH persist.
AdvGARCH-BestAsymmetric	EGARCH(1,1,1)	omega	0.0088	0.0056	0.116	0.9532	EGARCH persist.
AdvGARCH-BestAsymmetric	EGARCH(1,1,1)	alpha[1]	0.2353	0.0283	1e-16	0.9532	EGARCH persist.
AdvGARCH-BestAsymmetric	EGARCH(1,1,1)	gamma[1]	-0.0430	0.0189	0.0228	0.9532	EGARCH persist.
AdvGARCH-BestAsymmetric	EGARCH(1,1,1)	beta[1]	0.9532	0.0139	0	0.9532	EGARCH persist.

AdvGARCH-BestAsymmetric has a raw QLIKE comparison $p = 0.0015$ and Holm-adjusted $p = 0.0215$ within the final primary QLIKE family. AdvGARCH-BestQLIKE has raw $p = 0.0181$, but its Holm-adjusted $p = 0.2354$ does not support a decisive superiority claim after family-wise correction. HAR-Parkinson is close to GARCH(1,1), but the paired QLIKE difference is not statistically decisive. The statistical message is therefore class-level rather than single-winner: selected GARCH-family specifications and HAR-Parkinson are the strongest group.

D. Rolling-Origin Refit Robustness

Table IX compares static train-only forecasts with expanding-window monthly refits for the small preselected econometric set on the expanded common window. The conclusion largely survives. AdvGARCH-BestAsymmetric remains the best complete-window refit model, improving from static QLIKE 1.029992 to expanding-monthly QLIKE 1.023691. GARCH(1,1) improves from 1.062958 to 1.057908, and ARIMA-GARCH improves from 1.065995 to 1.059182. HAR-Parkinson remains stable, with QLIKE around 1.035.

The robustness experiment also reveals operational fragility. AdvGARCH-BestQLIKE is the primary validation-selected advanced specification in the static protocol, but its monthly expanding refit fails in 20 monthly segments on the expanded window and only produces 340 forecasts. The old 726-date baseline had 19 failed segments; the corrected 745-date win-

TABLE VIII
POST-FIT RESIDUAL DIAGNOSTICS FOR SELECTED GARCH-FAMILY MODELS.

Model	Test	Series	Lag	Statistic	p-value	Flag
GARCH(1,1)	Ljung-Box	Squared std. resid.	5	1.283	0.937	not_rejected
GARCH(1,1)	Ljung-Box	Squared std. resid.	10	3.417	0.97	not_rejected
GARCH(1,1)	Ljung-Box	Squared std. resid.	20	14.273	0.816	not_rejected
GARCH(1,1)	ARCH-LM	Std. resid.	5	1.267	0.938	not_rejected
GARCH(1,1)	ARCH-LM	Std. resid.	10	3.357	0.972	not_rejected
GARCH(1,1)	ARCH-LM	Std. resid.	20	13.936	0.834	not_rejected
AdvGARCH-BestQLIKE	Ljung-Box	Squared std. resid.	5	2.309	0.805	not_rejected
AdvGARCH-BestQLIKE	Ljung-Box	Squared std. resid.	10	3.789	0.956	not_rejected
AdvGARCH-BestQLIKE	Ljung-Box	Squared std. resid.	20	15.057	0.773	not_rejected
AdvGARCH-BestQLIKE	ARCH-LM	Std. resid.	5	2.273	0.81	not_rejected
AdvGARCH-BestQLIKE	ARCH-LM	Std. resid.	10	3.699	0.96	not_rejected
AdvGARCH-BestQLIKE	ARCH-LM	Std. resid.	20	14.517	0.803	not_rejected
AdvGARCH-BestAsymmetric	Ljung-Box	Squared std. resid.	5	1.938	0.858	not_rejected
AdvGARCH-BestAsymmetric	Ljung-Box	Squared std. resid.	10	3.249	0.975	not_rejected
AdvGARCH-BestAsymmetric	Ljung-Box	Squared std. resid.	20	15.089	0.771	not_rejected
AdvGARCH-BestAsymmetric	ARCH-LM	Std. resid.	5	1.921	0.86	not_rejected
AdvGARCH-BestAsymmetric	ARCH-LM	Std. resid.	10	3.213	0.976	not_rejected
AdvGARCH-BestAsymmetric	ARCH-LM	Std. resid.	20	14.528	0.803	not_rejected

TABLE IX
STATIC VERSUS EXPANDING-WINDOW MONTHLY REFIT ROBUSTNESS ON THE UNIFIED COMMON WINDOW.

Model	Protocol	N	Full	QLIKE	RMSE	MAE	Pred./Actual	Q90 QLIKE	Failed
Panel A: complete-window protocols									
ARIMA-GARCH	expanding_monthly	745	yes	1.059182	3.648986	1.432204	1.061	6.514	0
ARIMA-GARCH	static_train_only	745	yes	1.065995	3.651122	1.414546	1.032	6.762	0
AdvGARCH-BestAsymmetric	expanding_monthly	745	yes	1.023691	3.534515	1.357556	1.002	6.406	0
AdvGARCH-BestAsymmetric	static_train_only	745	yes	1.029992	3.543267	1.321160	0.938	6.728	0
AdvGARCH-BestQLIKE	static_train_only	745	yes	1.032548	3.561391	1.324605	0.936	6.756	0
EWMA(lambda=0.90)	expanding_monthly	745	yes	1.225435	3.696192	1.429298	1.019	8.345	0
EWMA(lambda=0.90)	static_train_only	745	yes	1.225435	3.696192	1.429298	1.019	8.345	0
GARCH(1,1)	expanding_monthly	745	yes	1.057908	3.635779	1.429309	1.066	6.516	0
GARCH(1,1)	static_train_only	745	yes	1.062958	3.628230	1.407866	1.036	6.758	0
HAR-Parkinson	expanding_monthly	745	yes	1.035385	3.623558	1.383026	1.015	6.114	0
HAR-Parkinson	static_train_only	745	yes	1.034521	3.597062	1.358346	0.976	6.335	0
Panel B: incomplete operational refits									
AdvGARCH-BestQLIKE	expanding_monthly	340	no	1.202280	4.350815	1.532191	0.927	7.762	20

dow adds a failed January 2023 segment. The incomplete refit has QLIKE 1.202280 and is marked as not covering the full common window. This does not invalidate the static primary result, but it weakens the practical attractiveness of using that ARIMA-EGARCH specification in a monthly operational refit setting.

Periodic updating improves some complete econometric forecasts and reduces the one-percent common-Normal VaR violation count for several models. For example, GARCH(1,1) declines from 16 to 15 one-percent violations after monthly refitting, while AdvGARCH-BestAsymmetric declines from 15 to 14. The changes are useful but do not remove the need for tail-risk diagnostics.

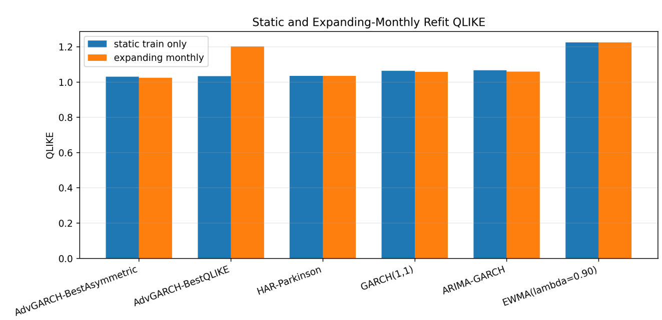

Fig. 4. Static and expanding-monthly QLIKE for the controlled refit robustness experiment.

E. Spike, VaR, and ES Diagnostics

Average QLIKE hides a common risk weakness. Table XI evaluates calm, high-volatility, and spike days using train/validation actual-variance thresholds. The strongest econometric models still underpredict most spike days, and severe underprediction rates remain high. HAR-Parkinson has spike-day QLIKE 8.821, while AdvGARCH-BestAsymmetric has 9.366. Corrected neural-dependent forecasts are not better overall, but some are better on spike-day QLIKE: ComboStack-PoolD_AllStable has 6.406 and ARIMA-GARCH-LSTM has 6.533. This supports a limited complementarity interpretation rather than a claim of neural dominance.

Table XII maps variance forecasts to VaR and ES using a common zero-mean Normal lower-tail return distribution. At the 5% and 2.5% levels, several econometric models have violation counts close to expectation. At the 1% level, strict-tail coverage is weaker and the expected count is only 7.45 observations. GARCH(1,1) has 16 one-percent violations with Kupiec $p = 0.0063$, while AdvGARCH-BestQLIKE and AdvGARCH-BestAsymmetric each have 15 with Kupiec $p = 0.0145$. For this sample size, 5% and 2.5% are more

TABLE X

FAILED EXPANDING-MONTHLY REFIT SEGMENTS RETAINED IN THE AUDIT TRAIL.

month	model	fit_rows	segment_rows	failure_stage
2023-01	AdvGARCH-BestQLIKE	3223	16	expanding_monthly_forecast
2023-02	AdvGARCH-BestQLIKE	3239	20	expanding_monthly_forecast
2023-06	AdvGARCH-BestQLIKE	3322	22	expanding_monthly_forecast
2023-09	AdvGARCH-BestQLIKE	3388	19	expanding_monthly_forecast
2023-10	AdvGARCH-BestQLIKE	3407	22	expanding_monthly_forecast
2023-11	AdvGARCH-BestQLIKE	3429	22	expanding_monthly_forecast
2023-12	AdvGARCH-BestQLIKE	3451	21	expanding_monthly_forecast
2024-02	AdvGARCH-BestQLIKE	3494	16	expanding_monthly_forecast
2024-03	AdvGARCH-BestQLIKE	3510	21	expanding_monthly_forecast
2024-04	AdvGARCH-BestQLIKE	3531	19	expanding_monthly_forecast
2024-06	AdvGARCH-BestQLIKE	3571	20	expanding_monthly_forecast
2024-08	AdvGARCH-BestQLIKE	3614	22	expanding_monthly_forecast
2024-09	AdvGARCH-BestQLIKE	3636	19	expanding_monthly_forecast
2024-10	AdvGARCH-BestQLIKE	3655	23	expanding_monthly_forecast
2024-11	AdvGARCH-BestQLIKE	3678	21	expanding_monthly_forecast
2024-12	AdvGARCH-BestQLIKE	3699	22	expanding_monthly_forecast
2025-01	AdvGARCH-BestQLIKE	3721	17	expanding_monthly_forecast
2025-02	AdvGARCH-BestQLIKE	3738	19	expanding_monthly_forecast
2025-03	AdvGARCH-BestQLIKE	3757	21	expanding_monthly_forecast
2025-10	AdvGARCH-BestQLIKE	3903	23	expanding_monthly_forecast

TABLE XI

CORRECTED-CONTEXT REGIME AND SPIKE DIAGNOSTICS.

model	overall_n	overall_qlike	calm_qlike	high_qlike	spike_n	spike_qlike	severe_spike_underprediction_rate
AdvGARCH-BestAsymmetric	745	1.029992	-0.075	4.379	45	9.366	0.889
AdvGARCH-BestQLIKE	745	1.032548	-0.077	4.393	45	9.391	0.889
HAR-Parkinson	745	1.034521	0.039	4.138	45	8.821	0.889
HAR-SquaredReturn	745	1.047975	0.039	4.204	45	8.939	0.889
GARCH(1,1)	745	1.062958	-0.038	4.403	45	9.406	0.867
RegimeAwareCombo-Vol20Q75	745	1.064346	0.277	3.535	45	7.350	0.889
ARIMA-GARCH	745	1.065995	-0.032	4.401	45	9.403	0.867
ComboStack-PoolD_AllStable	745	1.114856	0.451	3.200	45	6.406	0.867
Calibrated-Hybrid-LogTarget-Small-isotonic	745	1.126931	0.417	3.369	45	6.906	0.889
LSTM	745	1.163862	0.489	3.374	45	7.023	0.933
ARIMA-GARCH-LSTM	745	1.172986	0.539	3.223	45	6.533	0.911
EWMA(lambda=0.90)	745	1.225435	-0.146	5.335	45	11.971	0.889

TABLE XII

COMMON-NORMAL ZERO-MEAN VAR AND ES DIAGNOSTICS FOR SELECTED MODELS.

model	alpha	n_obs	var_violations	expected_violations	kupiec_uc_p_value	christoffersen_cc_p_value	quantile_loss
HAR-Parkinson	0.025	745	17	18.62	0.699	0.623	0.090158
RegimeAwareCombo-Vol20Q75	0.025	745	17	18.62	0.699	0.623	0.092862
GARCH(1,1)	0.025	745	22	18.62	0.441	0.682	0.092134
LSTM	0.025	745	15	18.62	0.379	0.399	0.096181
AdvGARCH-BestAsymmetric	0.025	745	24	18.62	0.227	0.467	0.090424
AdvGARCH-BestQLIKE	0.025	745	24	18.62	0.227	0.467	0.090618
ARIMA-GARCH-LSTM	0.025	745	13	18.62	0.163	0.3	0.096234
GARCH(1,1)	0.050	745	35	37.25	0.703	0.792	0.140396
AdvGARCH-BestAsymmetric	0.050	745	34	37.25	0.579	0.759	0.138450
AdvGARCH-BestQLIKE	0.050	745	34	37.25	0.579	0.759	0.138440
HAR-Parkinson	0.050	745	33	37.25	0.467	0.383	0.138978
RegimeAwareCombo-Vol20Q75	0.050	745	23	37.25	0.0102	0.0348	0.143732
LSTM	0.050	745	22	37.25	0.00566	0.02	0.149397
ARIMA-GARCH-LSTM	0.050	745	20	37.25	0.00153	0.00184	0.150913

informative for inference; 1% is treated as supplementary strict-tail evidence.

Table XIII adds a model-specific risk layer for selected GARCH-family models with verified Normal innovations. Because these selected specifications are Normal, their distribution-specific quantiles match the common-Normal zero-mean mapping. The validation-selected GED heavy-tail comparator is audited but not used for formal ES because GED ES parameter extraction and formula implementation were not verified in this pass.

F. Regime-Aware Dynamic Combination

The regime-aware combination is a controlled novelty extension, not a new model-search universe. The econometric expert and neural expert are selected by validation QLIKE, the regime indicator is origin-observable trailing 20-day volatility, the threshold grid is selected on validation data only, and low/high-regime weights are frozen before test evaluation. The selected design uses HAR-Parkinson and Calibrated-Hybrid-LogTarget-Small-isotonic, with a 75th-percentile validation threshold. It assigns econometric/neural weights of

TABLE XIII

DISTRIBUTION-SPECIFIC VAR AND ES FOR NORMAL GARCH-FAMILY SPECIFICATIONS.

model	alpha	n_obs	var_violations	expected_violations	kupiec_uc_p_value	christoffersen_cc_p_value	quantile_loss
AdvGARCH-BestAsymmetric	0.010	745	15	7.45	0.0145	0.0371	0.053518
AdvGARCH-BestQLIKE	0.010	745	15	7.45	0.0145	0.0371	0.053881
GARCH(1,1)	0.010	745	16	7.45	0.00631	0.0169	0.054979
GARCH(1,1)	0.025	745	22	18.62	0.441	0.682	0.092134
AdvGARCH-BestAsymmetric	0.025	745	24	18.62	0.227	0.467	0.090424
AdvGARCH-BestQLIKE	0.025	745	24	18.62	0.227	0.467	0.090618
GARCH(1,1)	0.050	745	35	37.25	0.703	0.792	0.140396
AdvGARCH-BestAsymmetric	0.050	745	34	37.25	0.579	0.759	0.138450
AdvGARCH-BestQLIKE	0.050	745	34	37.25	0.579	0.759	0.138440

TABLE XIV

VALIDATION-FROZEN STATIC AND REGIME-AWARE CORRECTED-NEURAL COMBINATIONS.

model	n_obs	qlike	rmse	mae	pred_actual_ratio
StaticCombo-HAR-Parkinson-with-Calibrated-Hybrid-LogTarget-Small-isotonic	745	1.077023	3.608901	1.520788	1.212
RegimeAwareCombo-Vol20Q75	745	1.064346	3.601104	1.486536	1.178

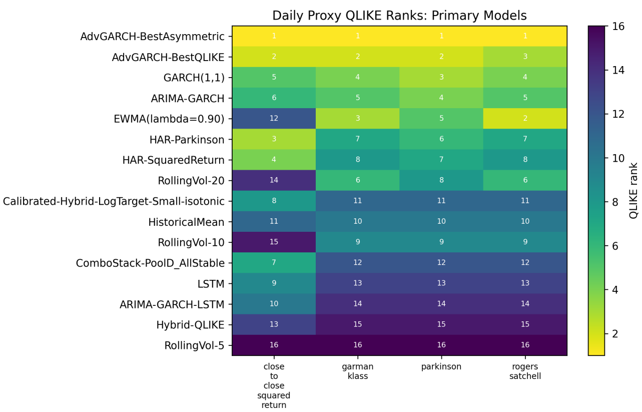

Fig. 5. Daily same-horizon proxy robustness for primary models. Numbers are QLIKE ranks; lower ranks are better.

0.45/0.55 in low regimes and 0.00/1.00 in high regimes. The regime-aware test QLIKE is 1.064346, better than the static convex combination at 1.077023 and close to GARCH(1,1), but still behind AdvGARCH-BestAsymmetric, AdvGARCH-BestQLIKE, and HAR-Parkinson overall. Its main contribution is spike behavior: spike-day QLIKE improves to 7.350. The result is useful as validation-safe evidence of high-volatility complementarity, not as a new primary winner.

G. Proxy Robustness

The proxy robustness exercise confirms that exact rankings are sensitive to how latent volatility is observed. Daily close-to-close, Parkinson, Garman-Klass, and Rogers-Satchell proxies measure related but non-identical volatility objects. The main conclusion remains class-level: GARCH-family specifications are consistently competitive, and HAR-Parkinson is strong on the squared-return target because it uses OHLC range information directly. Rolling Yang-Zhang sensitivities are kept separate because they are smoothed, overlapping measurements rather than direct one-day targets.

H. Discussion

The empirical findings are economically coherent. Raw VN-Index returns are heavy-tailed and negatively skewed, while squared returns exhibit strong dependence. GARCH-family models perform well because they encode this persistence directly. HAR-Parkinson performs competitively because daily range information carries volatility information that close-to-close squared returns alone may miss.

The neural and hybrid findings are best interpreted as modest and dataset-specific. Daily sample size is limited for deep sequence models, and smooth neural forecasts can look acceptable under some average losses while missing spike timing or tail risk. After corrected-context retraining, calibration and combinations can improve scale and spike behavior, but the implemented neural and hybrid approaches do not provide robust average-loss superiority over the strongest econometric benchmarks.

VI. LIMITATIONS

The study uses daily data only and lacks high-frequency realized volatility. Squared returns and OHLC range estimators are imperfect proxies for latent volatility, so proxy robustness is informative but not a substitute for realized-variance targets.

The advanced GARCH search is broad and numerically fragile. Although the static validation-selected models are useful, the refit experiment shows that AdvGARCH-BestQLIKE is not operationally stable under monthly expanding refits. The asymmetric comparator performs well, but it remains a category-selected robustness model rather than the sole primary winner.

Distribution-specific VaR/ES is implemented only for selected Normal GARCH-family models. GED and skewed-t ES are omitted because parameter extraction and ES formulas were not verified. Neural sequence construction was corrected and retrained into versioned output paths, but only a single deterministic seed was used for the full regenerated neural set. Random-initialization robustness therefore remains a limitation.

The dynamic-combination extension is deliberately small. Its validation-frozen regime-aware version improves spike behavior and beats the static combination, but it does not dominate the best econometric forecasts on overall QLIKE. It should therefore be read as a controlled robustness extension rather than a new primary forecasting model.

VII. Conclusion

This study provides a reproducible financial econometric evaluation of one-step-ahead daily VN-Index volatility forecasts. After corrected-context neural retraining, the leakage-safe unified common-window protocol expands to 745 target dates. Selected GARCH-family specifications and HAR-Parkinson remain the strongest group of forecasts. The best corrected-window QLIKE values are 1.029992 for the category-selected AdvGARCH-BestAsymmetric model, 1.032548 for the primary validation-selected AdvGARCH-BestQLIKE model, and 1.034521 for HAR-Parkinson.

The results are economically consistent with significant volatility clustering and persistent conditional variance in Vietnamese equity-index returns. The competitive performance of HAR-Parkinson indicates that daily OHLC range information is useful when high-frequency realized-volatility data are unavailable. The corrected neural and hybrid models do not robustly outperform the strongest econometric benchmarks in this daily-data setting, although calibrated and combined forecasts contain some high-volatility complementary information.

Risk diagnostics temper the average-loss findings. Spike underprediction remains common, and the 1% common-Normal VaR mapping rejects unconditional coverage for several strong average-loss models. The validation-frozen regime-aware combination improves over a static combination and lowers spike-day QLIKE, but it does not become the best overall forecast. The main lesson is therefore not that one model universally dominates. A credible daily VN-Index volatility forecast should be judged jointly by validation-based average loss, proxy robustness, operational refit stability, spike behavior, and tail-risk calibration.

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